AI DC SG-2

Week-2: The Transformer Architecture

<u>The Problem:</u>

In earlier natural language processing models, each word was given a single, fixed mathematical vector/embedding. This meant that the same word which could have multiple definitions, could not have had multiple representations. This led to a lot of ambiguity and became a blind spot for the models.

<u>The Solution:</u> (Self-Attention)

The Self-Attention mechanism solves that problem by inculcating the "context" that each word is being used in. This meant that the same word could now have different vector representations, based on its usage in a particular text.

<u>The Working Mechanism:</u>

The Transformer has layers called "Attention maps". These can be pictured as tabular forms that deal with 3 vectors that are created for each word/embedding:

1. <u>Query (Q)</u>: It "asks" for any contextual matching with all other words in the text
2. <u>Key (K):</u> It signifies what the word is offering to answer for a particular query
3. <u>Value (V):</u> It is the term to be added to the original embedding after being weighted by the attention values between the Query of that embedding and the Keys of all other words.

The values of the Query, Key, Value matrices that are used to produce these vectors are entirely learnt by the model during training. Their values can differ for each attention map/head.

The model computes the dot product of the Query and Key vectors of all possible combinations, masks any unnecessary words to prevent information leakage, then computes softmax of the remaining attention values for each Query vector. The Value vectors are then weighted by the corresponding softmax values and then summed up with the original embedding vector.

The new vector produced now has a contextual meaning to it, based on the other words in the text, which makes it far better than originally, and helps improve accuracy.

Example: Consider the word "apple" in these 2 sentences:

1. The farmer harvested a sweet, red apple.
2. The CEO unveiled the latest Apple iPhone.

The first sentence refers to the fruit apple, whereas the second one is referring to Apple Inc., the tech company.

In both cases, the word apple sends out a Query (Q) to each of the surrounding words, asking for the context that applies to it.
The preceding words respond with their Key (K) vectors. If the dot products are high, then the corresponding Value (V) of that word is added to the embedding of the original "apple" word after being scaled/weighed appropriately.

By the end of this process, the word has been mathematically altered to either be referring to the fruit, or the tech company because of the contextualization. The model will now have improved accuracy as the naïve errors of simple static representation have been avoided.